



TEAM SONAR · JOHANNESBURG

VUKA

You are not alone.

You don't have to ask.

VIGIL · the person **UMOJA** · the street **KHAYA** · the property **ANCHOR** · the record

Community safety you can verify.



Every section the brief asks for, in the order it asks.

1

Problem statement

Slides 3–6

Who exactly, and the evidence it is real

2

Solution

Slides 7–9

Four layers, and the one property no rival has

3

Technology readiness level

Slide 10

Assessed TRL 5, with what caps it there

4

Technological architecture

Slides 11–16

System, pipeline, trust boundary, chain, security

5

User journey story

Slides 17–18

The member's journey — and the subject's

6

Go-to-market strategy

Slides 19–21

Channel, unit economics, sustainability, impact

“Would a real user trust and use this?” Our answer is not that they should trust it. It is that they do not have to — they can check it.

You cannot open a bank account here without installing the app.

So all of us now carry the vault and the key in the same pocket — secured by a face or a finger that can be taken by force.

Criminals adapted before the banks did.

The robbery is no longer the wallet. It is: unlock the phone, open the app, transfer, then drive to an ATM for the daily limit.

The victim performs the transaction. Every record the bank holds says they authorised it — their face, their finger, their own device.



Nobody in a coerced robbery can reach for a phone and ask for help.

“Let someone call for help” was solved decades ago. All of it needs a free hand and a moment of privacy. Coercion removes both.

This is personal. A family member of this team was stopped at his own gate and told to unlock his phone. He got away because a neighbour happened to be looking.

Four failures. All the same shape.

1

Help requires a free hand

Panic buttons, apps, speed dials — every one needs a hand and privacy. Nothing on the market serves the person who cannot ask.

2

Alarms sense motion, not people

A sensor mounted high enough to ignore the dog is high enough for a person to crawl under — and cannot tell them apart anyway. So people stop arming it.

3

Nobody watches the watchers

Camera networks decide who is suspicious with no record a resident can inspect. The insider is the unmodelled threat across the whole category.

4

Whoever owns the log wins the dispute

A coerced transfer, a jammed remote, an argument over response time. The party holding the record has money riding on the outcome.

The problem is validated on real records,
not on a market-size slide.

15,712

INCIDENT RECORDS CLEANED

678

POLICE PRECINCTS JOINED

709

SUBURBS GEOCODED

9

PROVINCES COVERED

1,296

INCIDENTS IN THE 00:00 HOUR

3x

THE 05:00 TROUGH

680

HIJACKINGS IN THE SET

17 / 18

TARGET TOWNS SAPS COVERS

Crime here is concentrated in time and in place. Concentration is what makes it targetable — and it is why an 11–15 minute patrol dwell at the right hex at the right hour outperforms driving around all night.

Four users. Each one already pays for a worse version of this.

The commuter

Nomsa, 34, Tembisa → Midrand

Budget Android, prepaid data, drives home after dark. Owns a panic app she has never opened.

Needs help that starts without her hands.

The household

A family on one street in Kempton Park

Has an alarm they stopped arming because the dog sets it off. Pays R650/month for armed response.

Needs a sensor that can tell a person from a dog.

The control room

A mid-size security company, 40 vehicles

Fuel is the largest controllable cost. Response-time disputes are settled by their own log.

Needs a record the customer can check.

The subject

Anyone a camera saw — including the innocent

Has no way to learn what was recorded about them, or to have it removed.

Needs their own file, and a delete that holds.

The fourth user is the one nobody builds for. Every camera network has subjects. None of them gives a subject a way to see, check, or delete their own record. That is the gap we build into.

VUKA means wake up. It also stands for the four layers it is built from.



VIGIL

VIJ-il · Latin

THE PERSON

- Three independent senses on the phone
- Phone-to-car link breaking = hijack signal
- Covert duress PIN, unlocks normally
- Works offline, screen off, no data

Help that starts without her hands.



UMOJA

oo-MOH-ja · Swahili

THE STREET

- Learns who belongs before who does not
- Recurrence counts only if unknown
- Vectors, never images
- Risk layer · patrol optimisation

Whitelist before watchlist.



KHAYA

KY-ah · isiZulu

THE PROPERTY

- Tells a person from a dog
- Glass-break + gunshot, two-stage
- Deterrent siren when you are away
- 48–72h battery · LTE failover

An alarm you stop switching off.



ANCHOR

AN-ker · English

THE RECORD

- Hash chain over every decision
- Hourly root published publicly
- Two signatures to whitelist or disarm
- Any person can retrieve their file

Checkable by someone who distrusts us.

Every safety network here asks you to trust it. Ours can be checked instead.

The binding constraint on safety technology in South Africa is not accuracy. It is legitimacy. So every design decision is made to leave the system more checkable, not more capable.

A machine can never accuse you

ADR-0002

The machine's ceiling is `watch_candidate`. Only `human_verify()` with an operator ID sets `flagged`. Grep the repository — no other code path exists.

One operator cannot un-protect a street

ADR-0022

Whitelist, camera disarm, threshold change and deletion each need two independent signatures. Both are published.

You can hold your own file

ADR-0023

A subject can retrieve every decision made about them, with a proof anyone can verify without asking us.

The record was public before the dispute

ADR-0021

An hourly root, 32 bytes, on a chain we do not control. Precedence — not immutability.

The failures are published too

D14

Our forecast loses to a constant baseline and it is in the repository, on the honesty slide, and in this deck.

We are not proposing to build this. We shipped it in July.

33,600

LINES ACROSS THE REPOSITORIES

440

TESTS PASSING

25

ARCHITECTURE DECISION RECORDS

207

COMMITTS, DATED BEFORE THIS BRIEF

The commit history is the capability statement.

VUKA and its predecessor were built in July 2026 for a different programme. The CLAUDE.md in that repository, dated July, already describes a two-layer architecture that shipped. ANCHOR is the third layer of a plan we published before this hackathon existed.

- Kotlin foreground service, on-device INT8 inference
- FastAPI + WebSockets, migrations from day zero
- 15,712 records → 764 hotspot suburbs → 709 geocoded
- Calibration harness, false-alarm budget, held-out backtest
- SHA-256 evidence chain with a standalone verifier

Assessed TRL 5, with the evidence — and with what caps it there.

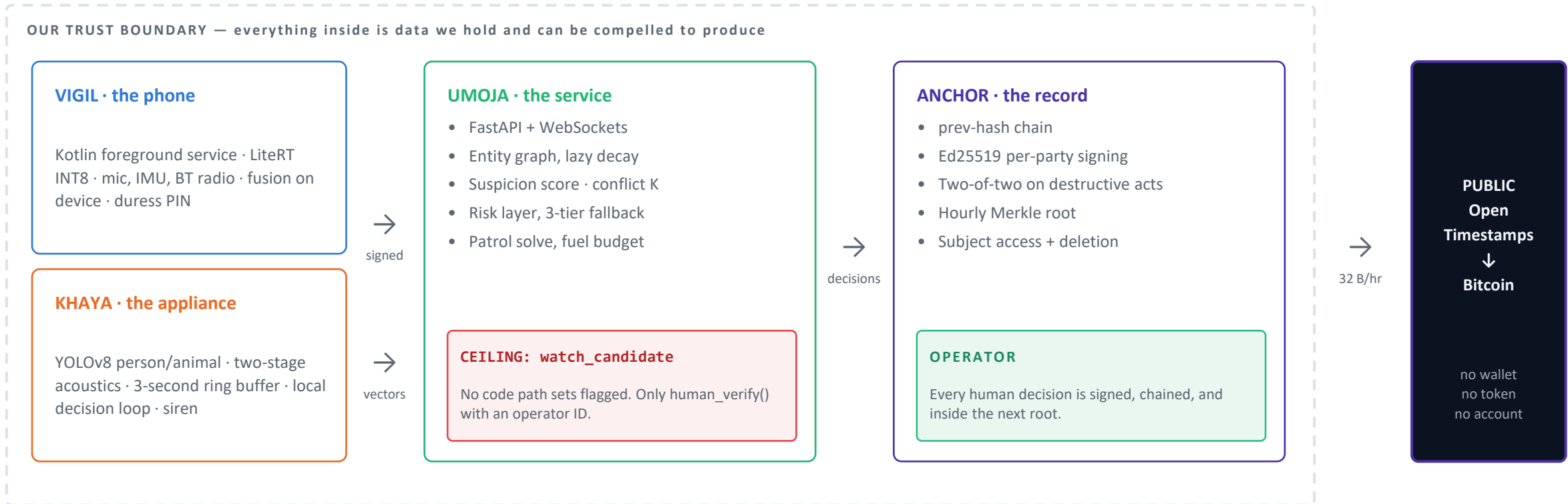
The template asks for TRL 3. We are stating our actual level, because understating it would misrepresent the work — and misrepresentation is the one thing our honesty ledger forbids.



● Integrated system running end to end across three machines over a public tunnel	TRL 5
● Latency measured under load in the intended topology, not simulated	TRL 5
● On-device model shipped inside an installable Android build	TRL 5
● Real data through the full pipeline: 15,712 records, 709 geocoded, 678 precincts	TRL 4–5
● 440 tests, including security-property tests on the human gate	TRL 4–5
● Appliance hardware not fabricated — breadboard and bench only	caps at 5
● No field deployment in real households yet	caps at 5

What moves us to TRL 6: a fabricated appliance in a real yard, for a season, with the false-alarm rate published whatever it says. That is the G-10 line on our public gap register.

Four layers, one boundary, and 32 bytes that leave.



Every layer states what it does when the layer above it disappears.

Phone offline: VIGIL still fuses and still records · Service unreachable: KHAYA still sirens and queues · Chain unreachable: the hash chain still verifies, the root publishes late and says so. Stale data is marked stale, never blank and never invented.

One incident, through the decision engine.

- SENSE** Three physically independent channels — acoustic, motion, vehicle link. Independence is the point: never stack correlated models and call it confidence.
- CALIBRATE** Each channel emits a calibrated probability. A raw softmax entering fusion is a correctness bug, not a style issue.
- FUSE** $\text{posterior} = \sigma(\text{prior} + \sum w_i \cdot \text{logit}(p_i))$. Log-odds, not Dempster–Shafer — D–S misbehaves under high conflict and our worst cases are high-conflict.
- CONFLICT GATE** Coefficient K rises when channels contradict. High K suppresses escalation and routes to verification. Disagreement is information, not noise.
- ESCALATE** L0→L3 with per-level hysteresis, so the state cannot chatter at a boundary. Every escalation carries a visible cancel window.

THE HARD BOUNDARY — ADR-0002, accepted July 2026

The machine's ceiling is watch_candidate. No code path can set flagged — only human_verify() can, and it requires an operator ID. A biometric match is a lead for a human, never a verdict.

then → human decides → signed record → hourly root published → suspicion decays at $0.5^{(\text{days}/7)}$

What crosses the boundary — and what never does.

DATA	LEAVES DEVICE	REACHES CLOUD	GOES ON CHAIN
Raw audio 3-second ring buffer, discarded. Only the label survives	✗ never	✗ never	✗ never
Face / plate images Vectors only — enforced in the vision layer, not in policy	✗ never	✗ never	✗ never
Embeddings (vectors) Needed for re-identification. Retention TTL in the data layer	✓ yes	✓ yes	✗ never
Location Public ledgers are permanent and correlatable	✓ yes	✓ yes	✗ never
Decision events Only the hash. The payload stays in our database	✓ yes	✓ yes	✓ hashed
Merkle root 32 bytes per hour, built server-side. A hash of a hash — the entire on-chain footprint	— n/a	✓ yes	✓ yes

“So you are putting biometrics on a blockchain?” No. We anchor a hash of a hash. Deleting a person’s data does not break the chain, because the chain never held it.

Blockchain does not make records immutable.

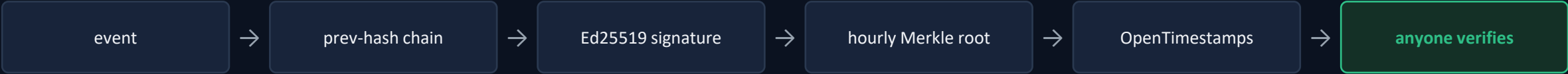
It proves they existed before you had a reason to lie.

The case a database cannot make

- 22:41 A duress signal is recorded on the phone
- 22:44 A transfer leaves the account
- 23:00 The hourly root is published — the 22:41 event is inside it
- 3 days later He disputes the transfer

The bank’s records show a correctly authenticated, correctly authorised instruction.

The anchored record proves the duress signal existed three minutes earlier — and was public before anyone knew there would be a dispute.



32bytes

ON CHAIN, PER HOUR

R1.30/month

ENTIRE NETWORK — FIXED, NOT PER USER

0

PERSONAL DATA ON CHAIN, EVER

Everyone worries about who a system puts on a watchlist. The dangerous power is who it takes off.

We found this by auditing our own code

```
def factor_f1_recurrence(entity, is_whitelisted):  
    if is_whitelisted:  
        return False    # exits immediately
```

A whitelisted entity accumulates no suspicion at all — and one operator could add one with a single unwitnessed API call.

The control — two of two, both published

- Whitelist an entity
- Disarm a camera
- Change a fusion threshold
- Delete a record

The attempt is itself an anchored event — including the one that fails.

● STRIDE model + abuse-case suite

Written before the code, run in CI

● Secret scanning, pre-commit and gate

After we committed secrets and disclosed it

● SAST · DAST · dependency scan

Gate G5, 24 September

● POPIA: retention TTL, subject access, deletion

Gate G3 — our own ADR recorded the gap first

Four blockchain features we designed — and rejected.



Alerts on chain

Our relay is 318 ms measured. The fastest chain is seconds. Consensus in the alert path costs time and buys nothing.



Autonomous gate unlock

A machine deciding alone to open someone's home. Violates ADR-0002 — and our own glass-break log holds a 0.71-confidence detection that was wrong.



First-responder bounties

Paying the first vehicle to arrive creates an incentive to race to a possibly-false alarm. We do not engineer a violent encounter.



Reward tokens

A database with an audit log beats a chain on cost, speed and UX. Tokenising it would be decoration.

Total autonomy in perception. Zero autonomy in consequence. The system may decide what it is looking at. It may never decide what happens to a person.

One incident. One clock. Three pairs of hands.

19:41	Nomsa's phone loses its link to her car while the car is still moving	<i>She touches nothing</i>
19:41	Three independent channels agree. Context lifts it: dark, a road with history. Conflict K stays low	<i>Threshold crossed → L2</i>
19:42	Her Guardian is notified. A countdown appears on her screen	<i>She can still stop this</i>
19:43	An operator opens the queue. The state reads watch_candidate — visibly locked there	<i>The machine cannot go further</i>
19:44	The operator dismisses it. A parking garage; the link dropped on the ramp	<i>Nothing was sent to police</i>
20:00	The dismissal is published inside the hourly root	<i>32 bytes</i>
later	Nomsa opens her record and verifies the dismissal happened and has not been altered	<i>So can a regulator</i>

The climax is 19:44 — the system decided not to accuse someone. 20:00 made that decision permanent.

The second journey belongs to the person the cameras saw.

Thabo delivers on that street twice a week. He was never a member, never asked, never consented to anything. Every camera network in the country has a Thabo. None of them gives him this.

1

He asks

GET /v1/subjects/{id}/record — every decision ever made about him: seen, scored, verified, dismissed, whitelisted. With actor IDs and timestamps.

2

He checks it himself

The bundle carries the chain entries, the Merkle path and the anchor reference. He can verify it with a public OpenTimestamps client that has never heard of us.

3

He asks for deletion

DELETE /v1/subjects/{id}/data — the payload is removed. The hash stays, so the chain still verifies. Deletion does not become a way to erase the fact that a decision was made.

4

He is forgotten anyway

Suspicion decays at $0.5^{(\text{days}/7)}$ with nobody required to clear him. Most systems need a human to decide to release you. Ours decides on its own to stop watching.

This is the POPIA answer, and it is also the product. A person can hold their own file without asking our permission to see it.

Sell through security companies, not against them.

They already hold the customer, the monthly billing rail and the vehicles on the road. Competing means buying households one at a time. Partnering means one integration reaches thousands — and customer acquisition cost collapses.

1 PROVE

GKSS chapters and our own street. 20–50 honest installs. Publish the false-alarm rate whatever it says.

2 PARTNER

One mid-size security company. Prove the fuel saving and provable response times on their own data.

3 BUNDLE

One short-term insurer. Their premium discount acquires the customer — we pay no marketing rand.

4 GOVERN

Estates and body corporates. One committee decision covers 200+ homes.

5 ANCHOR

A bank's fraud and disputes team. A verifiable duress record is unavailable today at any price.

6 CONTINENTAL

Kenya via Reboot Republic. Swap the crime dataset; the architecture ports.

Every adoption is self-interested — nobody is asked to do us a favour.

The member gets something that notices when they cannot reach a phone. The security company gets a lower fuel bill and proof it was fast when a customer says otherwise. The insurer gets evidenced claims. The estate gets a camera nobody can disable alone. The bank gets evidence that does not currently exist at any price.

A blockchain that gets cheaper per user as you grow.

KHAYA home bundle · R299 / month

Hardware amortised ($R3,000 \div 24$)	R125
Cloud + anchor, per household	R12
Support and operations	R25

Gross margin **R137 · 46%**

Prototype BOM R3,900, modelled at R3,000 at 1,000 units. Flagged as an estimate, not a quote.

Revenue lines

Security company licence	R35 / subscriber / mo
Patrol optimiser SaaS	R12,000 / control room
Estate / body corporate	R5,500 / estate / mo
Insurer verified installation	R40 / member / mo
Anchored claim record	R15 / record

R1.30 per month.

For the entire network.

720 anchors a month. Because every household is batched into one hourly root, it is a fixed cost — at a hundred homes or a hundred thousand, the same number.

Most blockchain products die on cost and UX. Ours has no wallet, no token, no per-user fee, and no personal data on chain.

What keeps running after demo day — and how you would measure it.

SUSTAINABILITY — why it survives

- Anchoring cost is fixed, not per user — growth cannot break the unit economics
- No token, no wallet, no exchange listing — nothing to collapse
- Runs on a laptop and a tunnel; cloud is a flex, never the decision path
- MIT-licensed core, public gap register — someone else can carry it if we stop
- Chosen for longevity: a claim may reach a court in ten years

MEASURABLE IMPACT — the metrics we publish

False alerts per camera-week	≤ 1
Detection → alert render, p95	≤ 2,000 ms
Anchor liveness — hours with a root	100%
Dual-signature compliance	100%
Subject requests answered inside 30 days	100%

G2 · 8 Sep

Anchored

Merkle batching, Ed25519 signing, public anchor, verifier extended

G3 · 14 Sep

Governed

Two-signature gates, subject access, deletion route — POPIA gap closed

G4 · 20 Sep

Calibrated

Fusion weights fitted on real data. Reliability diagram and ECE published

G5 · 24 Sep

Hardened

SAST, DAST, dependency scan, abuse-case suite, ingest fuzzing

The dates are gates, not hopes. Each one has a test that has to pass before the next begins, and the build log is append-only.



Seven builders, four universities. Contracts frozen before implementation.

LH

Lethabo Hoaeane

University of South Africa

PROFILER · CO-LEAD

Architecture, product and UX. The spec, the ADRs, final review.

SK

Sibusiso Khumalo

Witwatersrand

BACKEND · CO-LEAD

Service, evidence chain, CI, demo orchestration.

BA

Babatunde Adelusi

Pretoria

BUSINESS DEVELOPER

Business case, unit economics, go-to-market, the pitch.

MC

Mutarisi Chibaya

Pretoria

FRONTEND DEVELOPER

Operator dashboard and the member-facing screens.

KM

Khutso Mothopa

Witwatersrand

SYSTEM ANALYST

Requirements, work breakdown, traceability, verification map.

VK

Vukosi Khoza

Witwatersrand

IOT DEVELOPER

KHAYA appliance — sensors, edge runtime, power, tamper.

IM

Ipeleng Constance Modise

Tshwane University of Technology

SECURITY DESIGNER

OWASP coverage, threat model, SSDLC, physical security.

Skills coverage

Software · backend · enterprise architecture · UI/UX · IoT and hardware · security design · systems analysis · business development. Four universities, and gender variety in the team.

Stated gap

We are distributed across four universities and will not have built together in one room until the 25th. UI/UX is now owned rather than shared — but by a co-lead who also owns architecture, which is a single point of failure we are managing, not one we have removed.

The numbers that embarrass us, and the claims we refuse.

Open gaps — disclosed, not discovered

- Our forecast loses to a constant baseline: MAE 0.484 vs 0.246. Published
- Fusion weights are hand-set, not fitted — the params file says so in machine-readable text
- Face thresholds 0.55 / 0.65 cosine are targets. Uncalibrated, and labelled as targets
- Latency is $n = 10$. A real p95 needs a real sample
- Bias evaluation on the face pipeline has not been run yet

Claims we refuse to make

- “Identifies criminals” — it produces leads; humans identify
- “Court-admissible” — ours is structured to support a case
- “Unbiased AI” — bias-aware, and the human gate is the answer
- “Unhackable” — nothing is. Our claim is that intrusion is visible
- “Prevents crime”, and any uncalibrated precision, ever

Why we publish the failures

One of our own scripts opens by attacking its own project — “that is descriptive analytics, not prediction, no matter how it is presented” — and then gives the reader three ways to catch it lying. A parameter file carries a machine-readable warning against our own future overclaiming. A competitor can copy the practice. They cannot produce six weeks of it, dated before they knew they would need it.

It forgets you.

It never forgets what it did.

A documented seven-day half-life, with nobody required to clear you. Most surveillance systems need a human to decide to release you. Ours decides on its own to stop watching.

Every safety network in South Africa asks you to trust it. Ours can be verified.

VUKA · You are not alone. You don't have to ask.

